

DIGITAL IMAGE PROCESSING

Semester Project Document

General Instructions:

- This is an individual project.
- Students are required to complete the assignment using **Octave or Anaconda (Python with Jupyter Notebook)**.
- Direct copy-paste without understanding is strictly not allowed.

Project Guidelines:

Input Image:

- Select any digital image (preferably a noisy image).
- Ensure the image is clear and suitable for processing tasks

Implementation Steps:

Students must perform the following tasks in sequence:

1. Import the image into the software
2. Apply or use a noisy image
3. Perform noise removal using suitable filtering technique
4. Convert the image from RGB to Grayscale
5. Apply image enhancement techniques (filtering)
6. Perform basic object detection on the image

Output Requirements:

- Each step must generate and display a separate output.
- Before and after results must be clearly shown.

Video Demonstration:

- Students must record a **short video (3–5 minutes)**.
- It is mandatory in case of any emergency you should have evidence of your work.
- The video should include:
 - Step-by-step code execution
 - Display of outputs

Submission Requirements:

- Source code file (.py or .m)
- Input and output images
- Video demonstration file

. Evaluation Criteria:

- Correct implementation of all tasks
- Quality of results
- Understanding shown in video presentation
- Proper documentation and formatting

Important Note:

- Late submissions will not be accepted.
- Plagiarism will result in deduction of marks.

Marks ditribution:

- Project submission=50
- Project viva=50

Assigned by :HAFSA HAFEEZ

BEST OF LUCK:)